Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 04)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

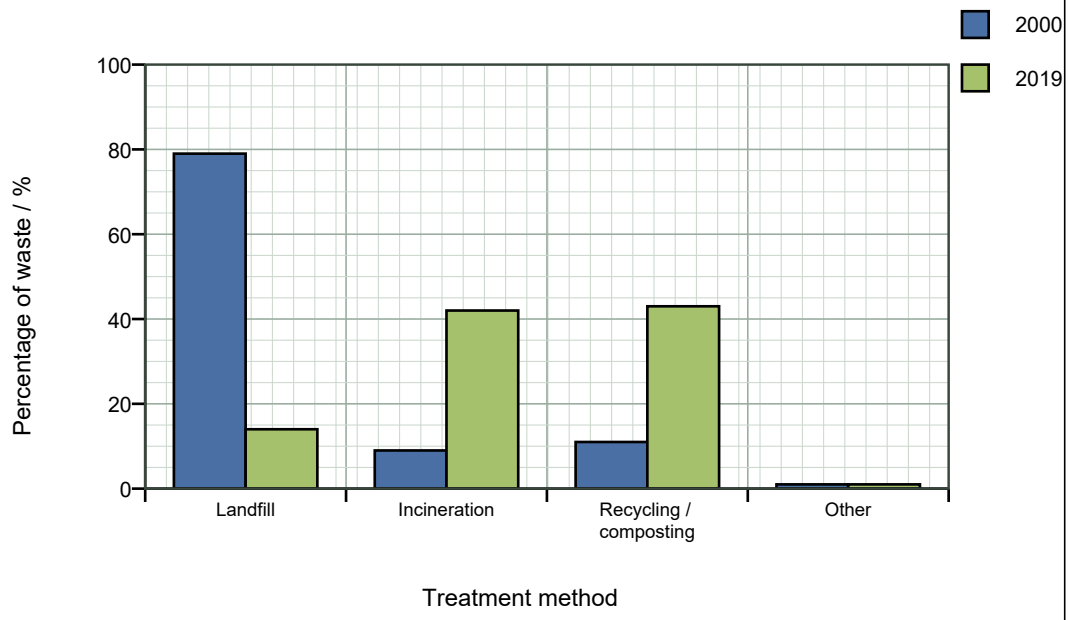
Question	Mark
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TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 1 . 1 Describe the changes shown in Figure 1.

[2 marks]

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0 1 . 2 Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

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0 1 . 3 Describe **four** features of a well-managed landfill site.

[4 marks]

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0 1 . 4 Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

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0 2 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 2 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 2 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 3 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

0 3 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer _____ times

0 3 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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0 3 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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0 3 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

0 3 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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0 4 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 4 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 4 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 5 Table 1 shows the approximate energy density of some energy resources.

Table 1

Energy resource	Energy density / MJ kg^{-1}
Uranium (natural, in fission reactor)	500 000
Natural gas	55
Petrol	46
Coal	28
Dry wood	16
Lithium-ion battery (stored electricity)	0.7

0 5 . 1 Explain what is meant by **energy density** and why it is an important feature of an energy resource.

[3 marks]

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0 5 . 2 Use Table 1 to calculate how many tonnes of coal contain the same energy as 1 kg of uranium.

[2 marks]

Answer ☐ tonnes

0 5 . 3 Explain why low energy density does not prevent solar and wind power from being useful.

[2 marks]

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0 5 . 4 Explain the difference between an **intermittent** resource and an **unpredictable** resource, using named examples.

[2 marks]

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0 5 . 5 Explain why natural gas must be liquefied for transport by ship.

[1 mark]

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0 6 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 6 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 6 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 7 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

0 7 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

Properties causing ozone depletion

0 7 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 7 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 7 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 7 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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0 8 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 8 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 8 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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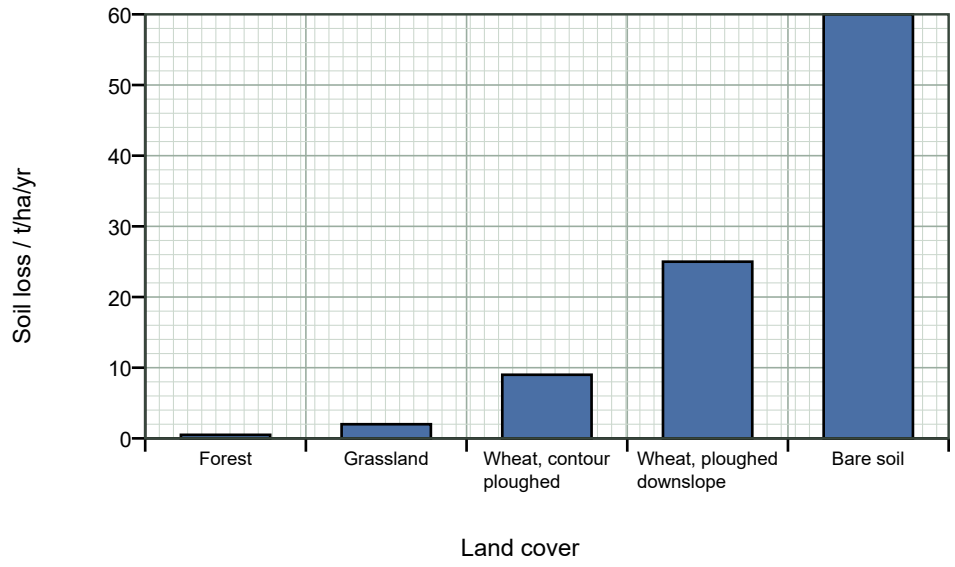
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0 9 Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

Figure 1



0 9 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 9 . 2 Explain **four** ways in which vegetation reduces the rate of soil erosion.

[4 marks]

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0 9 . 3 Explain why ploughing up and down a slope causes more erosion than contour ploughing.

[2 marks]

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0 9 . 4 Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

[2 marks]

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1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

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1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

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1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS